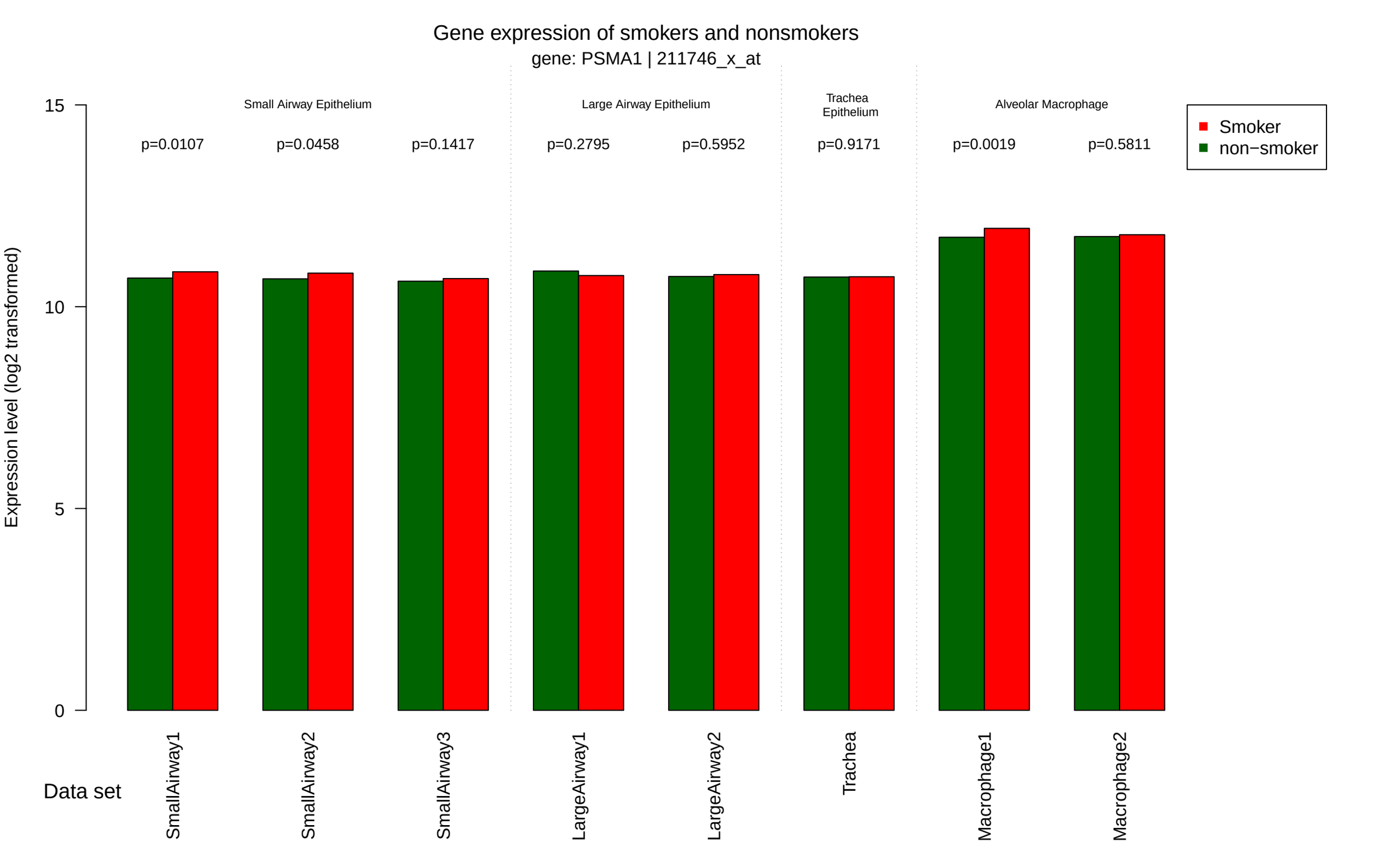
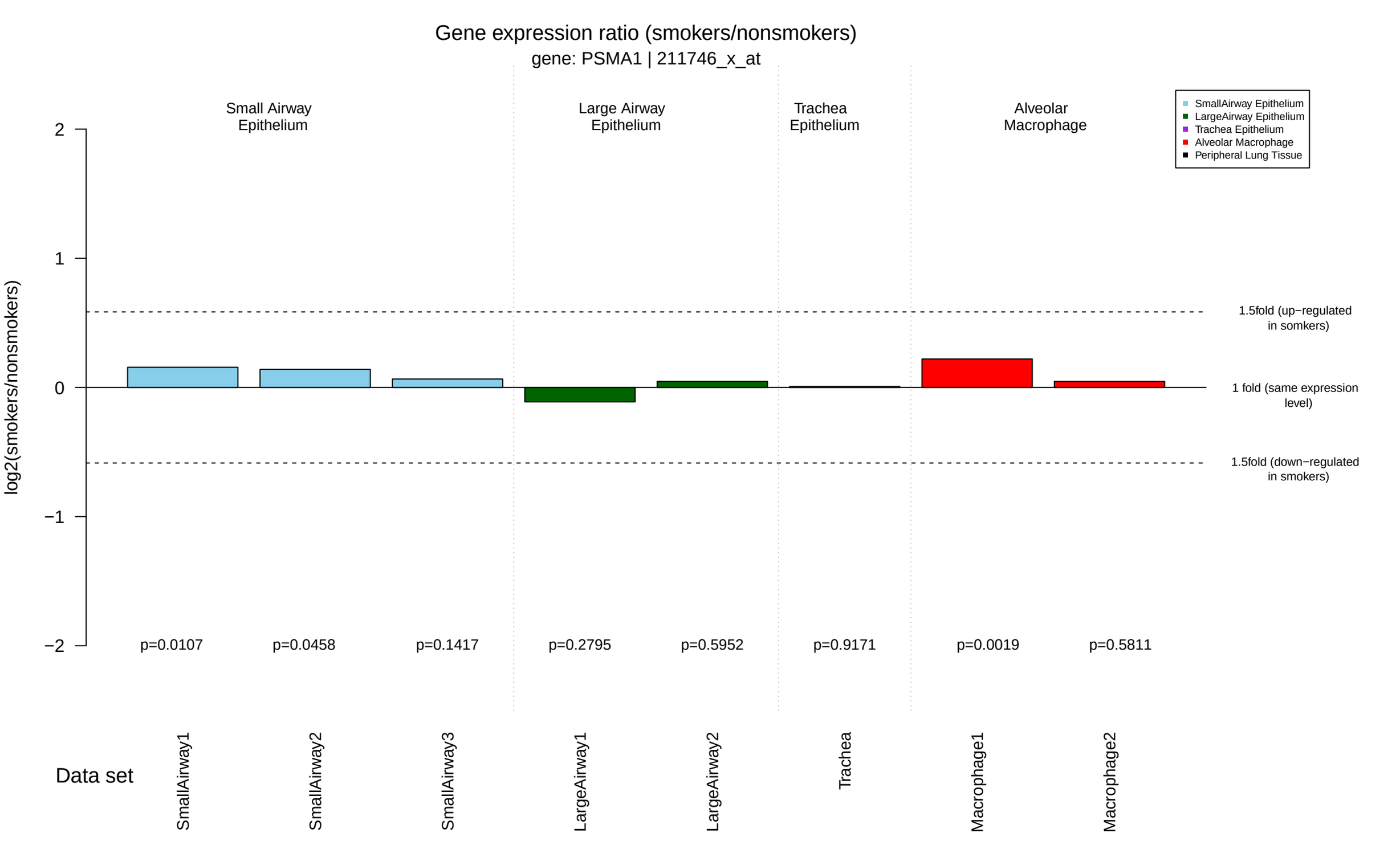
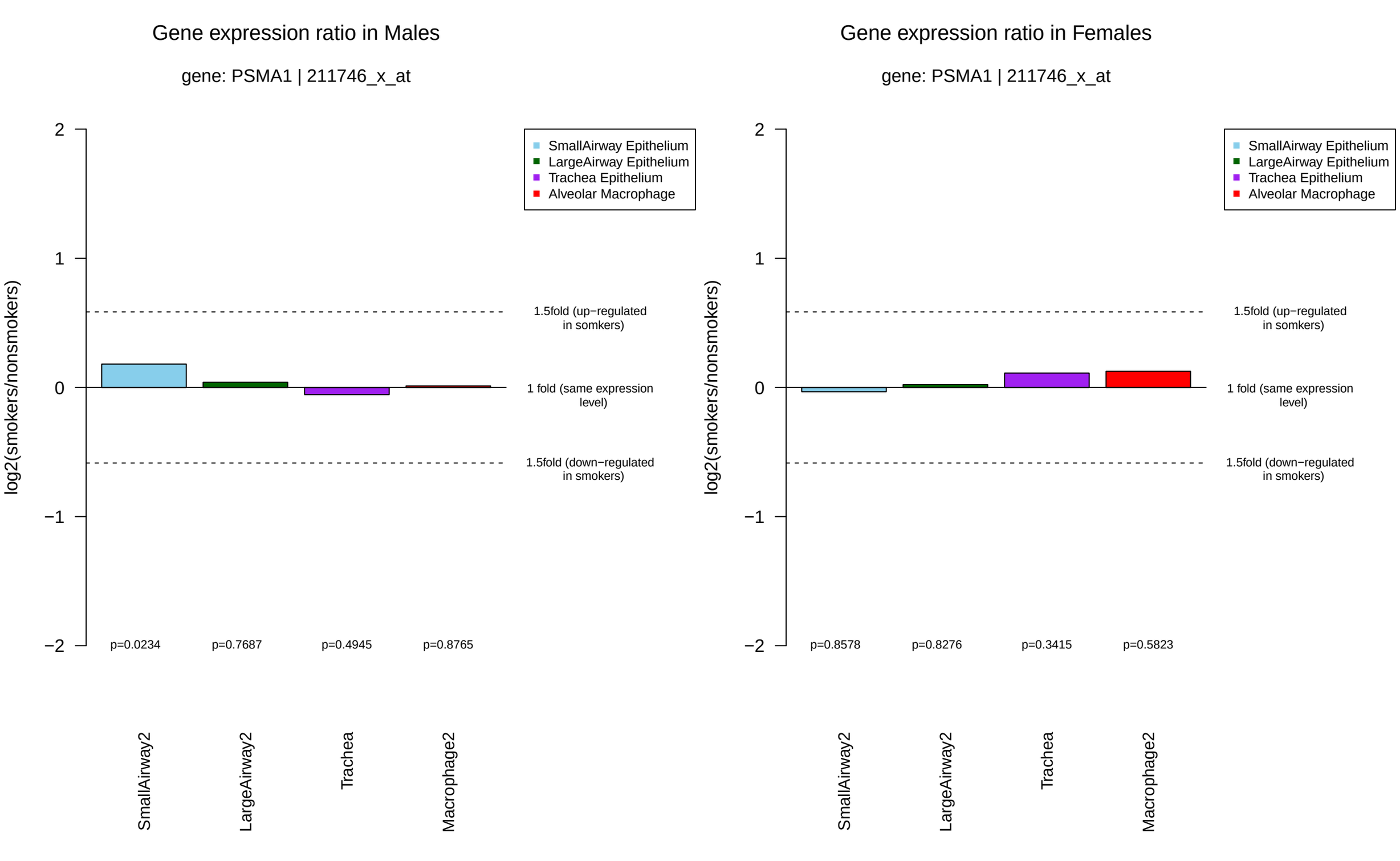
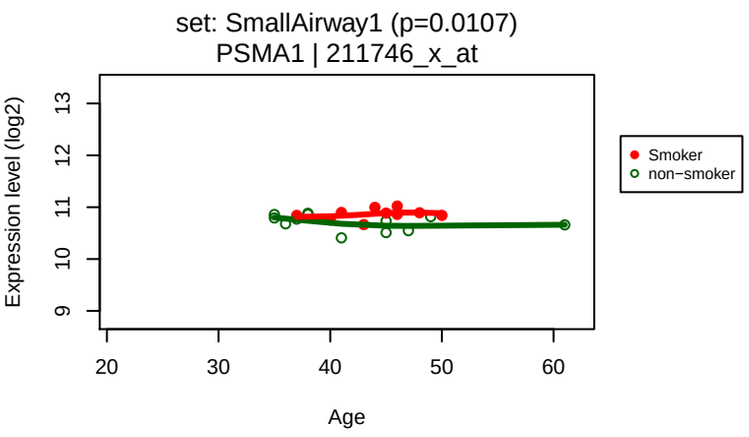
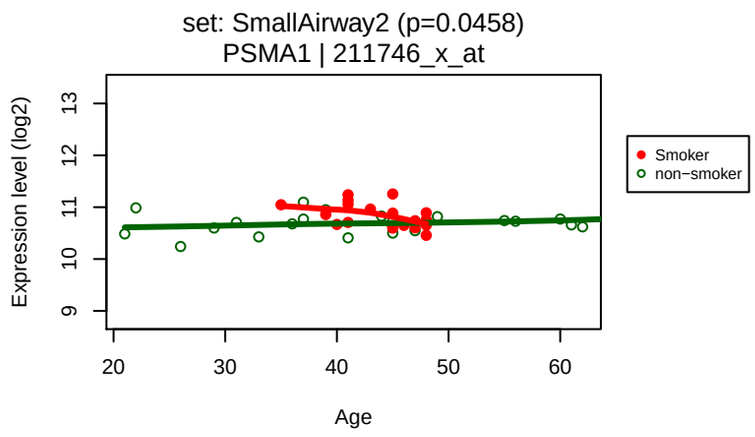
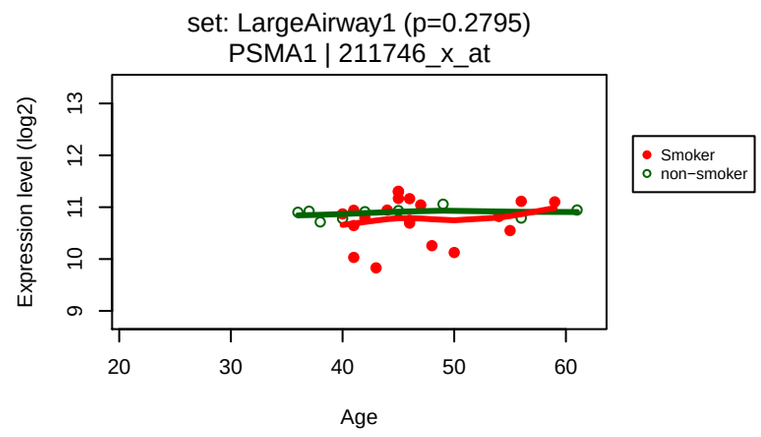

Gene expression of smokers and nonsmokers
gene: PSMA1 | 211746_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0107
p=0.0458
p=0.1417
p=0.2795
p=0.5952
p=0.9171
p=0.0019
p=0.5811
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

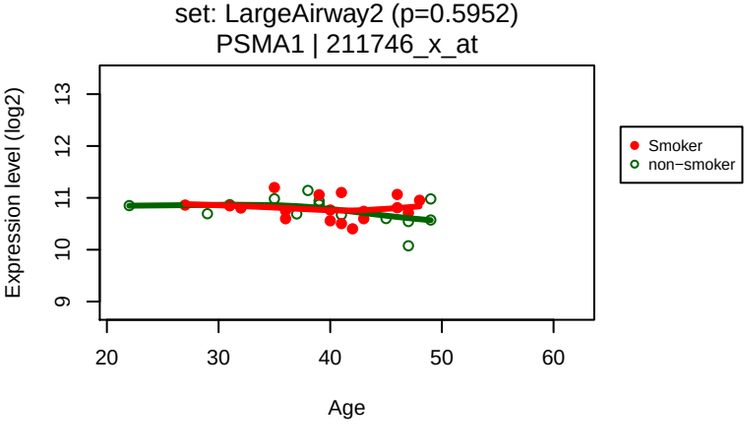

Gene expression ratio (smokers/nonsmokers)
gene: PSMA1 | 211746_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0107
p=0.0458
p=0.1417
p=0.2795
p=0.5952
p=0.9171
p=0.0019
p=0.5811
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

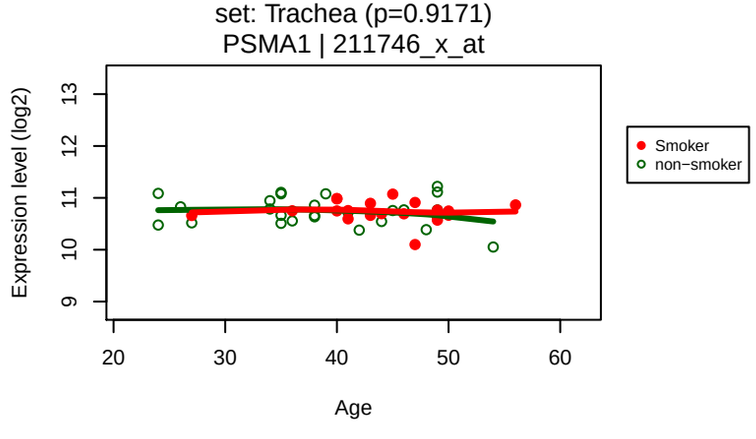

Gene expression ratio in Males
gene: PSMA1 | 211746_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0234
p=0.7687
p=0.4945
p=0.8765
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA1 | 211746_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8578
p=0.8276
p=0.3415
p=0.5823
SmallAirway2
LargeAirway2
Trachea
Macrophage2

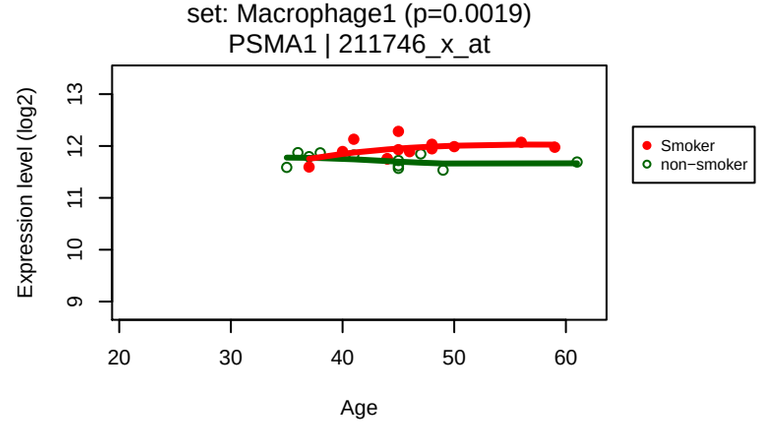

set: SmallAirway1 (p=0.0107)
PSMA1 | 211746_x_at
Expression level (log2)
Age
Smoker
non−smoker

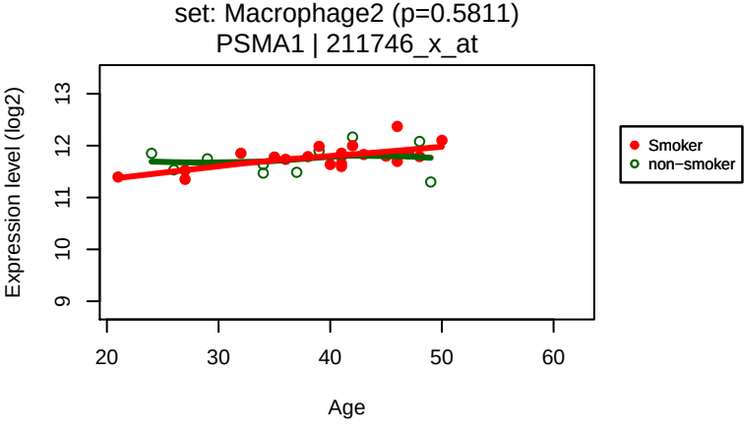

set: SmallAirway2 (p=0.0458)
PSMA1 | 211746_x_at
Expression level (log2)
Age
Smoker
non−smoker

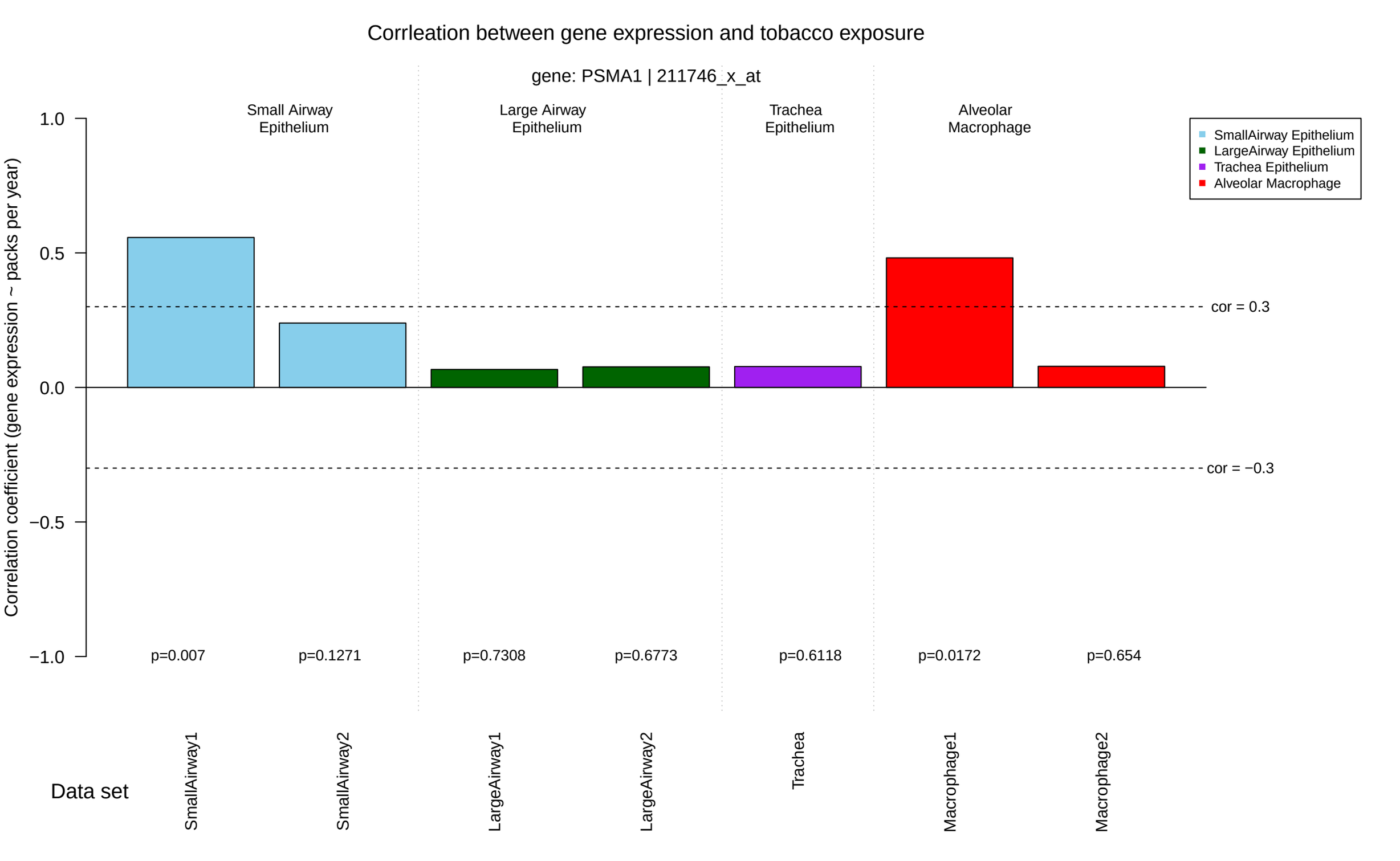

set: LargeAirway1 (p=0.2795)
PSMA1 | 211746_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5952)
PSMA1 | 211746_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9171)
PSMA1 | 211746_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0019)
PSMA1 | 211746_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5811)
PSMA1 | 211746_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA1 | 211746_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.007
p=0.1271
p=0.7308
p=0.6773
p=0.6118
p=0.0172
p=0.654
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2